Module 1: Advanced Linear Algebra

Previous Year University Examination Questions

PART A – Short Answer Questions (2–5 Marks)

- 1. Define singular and non-singular matrices.
- 2. State necessary and sufficient condition for invertibility of a matrix.
- 3. State the Rank–Nullity Theorem.
- 4. Define kernel and image of a linear transformation.
- 5. What is meant by full rank of a matrix?
- 6. When does a homogeneous system have non-trivial solutions?
- 7. Define dimension of a vector space.

PART B – Problem Solving Questions (5–8 Marks)

- 8. Show that if $\det(A) \neq 0$, then A is invertible.
- 9. Determine whether a given matrix is singular or non-singular using rank method.
- 10. Verify the Rank–Nullity Theorem for a given matrix.
- 11. Prove that columns of an invertible matrix are linearly independent.
- 12. Determine whether a given system $Ax = b$ is consistent using rank condition.
- 13. Show that if $Ax = 0$ has non-trivial solution, then A is singular.

PART C – Long Answer / Theory Questions (8–12 Marks)

- 14. Prove that a square matrix is invertible if and only if its determinant is non-zero.
- 15. Explain the relationship between rank, nullity, kernel, and image.
- 16. Prove that eigenvectors corresponding to distinct eigenvalues are linearly independent.
- 17. Show that if eigenvectors form a basis, the matrix is diagonalizable.
- 18. Discuss the behavior of the system $Ax = b$ when A is singular.
- 19. Prove that if $\text{rank}(A) = n$ for an $n \times n$ matrix, then A is non-singular.